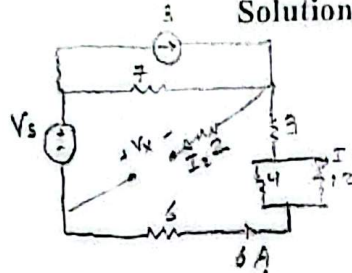




Q1:

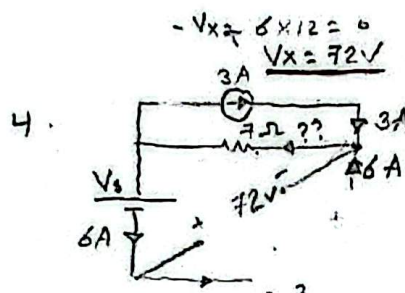
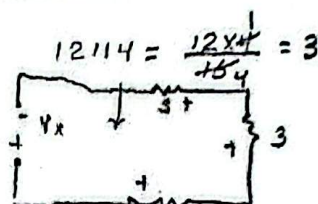
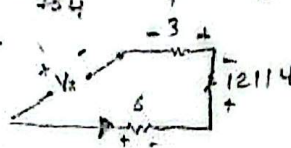
Solutions:



$$1. I_1 = - \frac{6 \times 4}{4+3} = - \frac{6}{4} = - \frac{3}{2} = -1.5A$$

$$2. I_2 = 0$$

$$3. V_x \rightarrow$$



$$6+3=9A$$

$$7 \times 9 = 63V$$

$$\frac{63}{1.5} = 42V$$

$$5. P_s = -135 \times 6 = -810W \checkmark$$

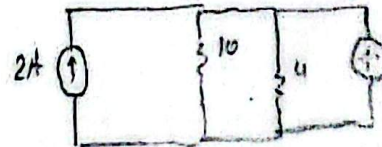
$$6. P_{2\Omega} = 0$$

Q2:

$$1. V_1 = -8V \checkmark$$

$$2. V_2 = -20V \checkmark$$

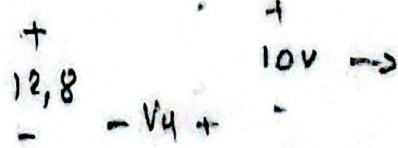
$$3. V_3 \rightarrow$$



$$I_{10\Omega} = \frac{4.5 \times 4}{14} = 1.28A$$

$$V_{10\Omega} = V_3 = 1.28 \times 10 = 12.8V \checkmark$$

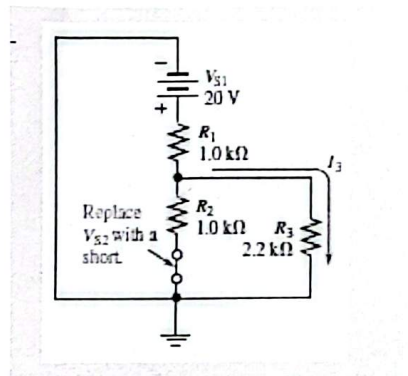
$$4. V_4 \rightarrow$$



$$12.8 - V_4 - 10 = 0$$

$$V_4 = 12.8 - 10 = 2.8V \checkmark$$

$$5. P_{6\Omega} = \frac{12 \times 12}{6} = \frac{144}{6} = 24W \checkmark$$



Looking from V_{S1} ,

$$R_{T(S1)} = R_1 + \frac{R_2 R_3}{R_2 + R_3} = 1.0 \text{ k}\Omega + \frac{(1.0 \text{ k}\Omega)(2.2 \text{ k}\Omega)}{3.2 \text{ k}\Omega} = 1.69 \text{ k}\Omega$$

$$I_{T(S1)} = \frac{V_{S1}}{R_{T(S1)}} = \frac{20 \text{ V}}{1.69 \text{ k}\Omega} = 11.8 \text{ mA}$$

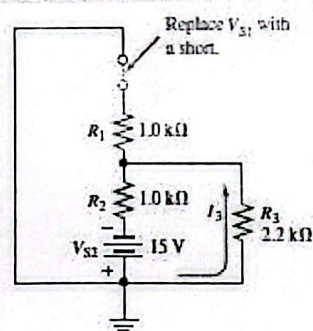
Now apply the current-divider formula to get the current through R_3 due to source V_{S1} .

$$I_{3(S1)} = \left(\frac{R_2}{R_2 + R_3} \right) I_{T(S1)} = \left(\frac{1.0 \text{ k}\Omega}{3.2 \text{ k}\Omega} \right) 11.8 \text{ mA} = 3.69 \text{ mA}$$

Notice that this current is downward through R_3 .

Step 2: Find I_3 due to source V_{S2} by replacing source V_{S1} with a short, as shown in Figure 28.

FIGURE 28



Looking from V_{S2} ,

$$R_{T(S2)} = R_2 + \frac{R_1 R_3}{R_1 + R_3} = 1.0 \text{ k}\Omega + \frac{(1.0 \text{ k}\Omega)(2.2 \text{ k}\Omega)}{3.2 \text{ k}\Omega} = 1.69 \text{ k}\Omega$$

$$I_{T(S2)} = \frac{V_{S2}}{R_{T(S2)}} = \frac{15 \text{ V}}{1.69 \text{ k}\Omega} = 8.88 \text{ mA}$$

Now apply the current-divider formula to find the current through R_3 due to source V_{S2} .

$$I_{3(S2)} = \left(\frac{R_1}{R_1 + R_3} \right) I_{T(S2)} = \left(\frac{1.0 \text{ k}\Omega}{3.2 \text{ k}\Omega} \right) 8.88 \text{ mA} = 2.78 \text{ mA}$$

Notice that this current is upward through R_3 .

Step 3: Calculate the total current through R_3 .

$$I_{3(\text{tot})} = I_{3(S1)} - I_{3(S2)} = 3.69 \text{ mA} - 2.78 \text{ mA} = 0.91 \text{ mA} = 910 \mu\text{A}$$

This current is downward through R_3 .